Music Education Platform

Development Brief

Client Name:
Tanveer Haque

Project Overview

Foundations of Music is a bilingual online music theory course created by Tanveer Haque, an independent musician and music educator based in Dhaka, Bangladesh.

The initial course consists of **9 modules** covering fundamental music theory, including sound, notation, rhythm, intervals, keys, the circle of fifths, and major/minor scales.

The course content is complete, including written material, exercises, quizzes, slides, audio, notation graphics, and videos.

The platform will initially sell this course directly to students, with the ability to add additional courses in the future.

The primary market is Bangladesh, while international students and international payments should also be supported.

Project Objective

Build a **simple, professional, secure and maintainable online learning platform** through which students can:

- Discover the course
- Review the curriculum
- Purchase the course
- Create an account
- Access the course after successful payment
- Watch lessons
- Study written/audio/visual materials
- Complete quizzes and exercises
- Track their progress
- Return later and continue their studies

The platform should feel like a high-quality educational product rather than a complex technology platform.

Priorities:

Reliability → Security → Simplicity → Maintainability → User experience → Aesthetics → Advanced features

Required Student Experience

Public Website:

- Home
- About/instructor
- Course landing page
- Curriculum
- Pricing
- FAQ
- Contact
- Login
- Registration
- Terms & Conditions
- Privacy Policy
- Refund Policy

Required Student Experience

Student Account:

Students must be able to:

- Register
- Log in/out
- Reset their password
- Manage basic account information
- View purchased courses
- Access lessons
- See overall course progress

Course Structure

Each course should support:
Course → Modules

Each module contains:

- MP4 video
- Written explanations in Markdown
- MP3 audio
- SVG notation/diagrams
- Downloadable resources
- Multiple-choice quizzes

Quiz System

Quizzes contain:

- Multiple-choice questions
- 4 answer options per question
- Some questions containing SVG diagrams and/or MP3 audio

For the initial course, a module remains locked until the student completes its quiz with **100% accuracy**.

Students may retry questions until they select the correct answer. The correct answer is revealed when selected.

The developer should recommend the simplest reliable implementation of this system.

Payments

The initial market is Bangladesh.

The platform should support:

- BDT pricing
- Visa
- MasterCard
- bKash
- Appropriate local payment methods
- International payments where practical
- Automatic payment verification
- Automatic enrollment after successful payment
- Purchase/order records
- Payment confirmation

Payments

Potential providers include **SSLCOMMERZ** and **aamarPay**, but the developer should recommend the most appropriate current solution based on:

- Availability
- Fees
- Reliability
- API/integration quality
- Settlement process
- International payment support
- Requirements for a Bangladesh-based business

DECISION REQUIRED: Final payment provider.

Video Delivery

Course videos are commercially valuable content.

Videos should **not** simply be exposed as publicly accessible MP4 files.

The developer should recommend an appropriate secure video-streaming solution, potentially including Cloudflare Stream, using mechanisms such as:

- Signed URLs/tokens
- Access controls
- Domain restrictions
- Authentication/authorization

The objective is to prevent unauthorized access and casual redistribution, not to make screen recording technically impossible.

Original video masters will remain separately archived by the course owner.

DECISION REQUIRED: Final video-hosting/streaming provider.

Content Management

The administrator should be able to manage ordinary course content without requiring a developer for every change.

The system should ideally allow the administrator to:

- Add downloadable resources
- Create/edit quizzes
- Modify course descriptions
- View students
- View enrollments
- View purchases
- Manage student access
- Create/edit courses
- Create/edit modules
- Create/edit lessons
- Edit lesson text
- Add videos
- Add audio
- Add images/SVGs

The developer should clearly identify which functions will be available through the proposed CMS/admin interface.

Future Expansion

The platform should be capable of supporting additional courses without requiring a complete rebuild.

Future possibilities include:

- Multiple courses
- Course categories
- Additional educational materials
- Additional quizzes
- Certificates
- Coupons/discounts
- Additional instructors

These do **not** all need to be implemented in Version 1.

A second part of the Core Music Theory course is already being developed. It should eventually be available either:

- Only to students who completed Part 1, or
- As a combined Part 1 + Part 2 bundle.

The architecture should accommodate this progression/bundling model.

Design Direction

The website should be:

- Simple
- Clean
- Modern
- Professional
- Educational
- Calm
- Highly readable
- Mobile responsive

Avoid:

- Excessive animation
- Unnecessary visual effects
- Complicated dashboards
- Excessive gradients
- Generic "tech startup" aesthetics
- Unnecessary features

The educational content should remain the focus.

Technical Approach

The developer should recommend the most appropriate architecture rather than assuming that custom development is necessary.

The following approaches should be considered:

1) WordPress + LMS plugin
2) Hosted LMS
3) Custom web application
4) Hybrid solution

Factor	**Requirement**
Security	High
Maintainability	High
Ease of content management	High
Performance	High
Ownership	High
Flexibility	High
Scalability	Moderate–High
Initial complexity	Keep low
Developer dependency	Keep reasonable

The proposed architecture should address:

- **Frontend**
- **Backend**
- **Database**
- **Authentication**
- **CMS/LMS**
- **Video hosting**
- **Payment integration**
- **Email**
- **Analytics**
- **Hosting**
- **CDN**
- **DNS**
- **HTTPS/SSL**
- **Backups**
- **Security**
- **Monitoring**
- **Maintenance**

The developer should explain important technical choices in language understandable to a non-developer.

Security

The developer must explain their approach to:

- Authentication
- Password security
- Authorization
- Student data protection
- Payment security
- Video access control
- HTTPS
- Database security
- Server security
- Rate limiting
- Input validation
- Software/plugin updates
- Backups and recovery
- Admin security
- API-key/secrets management

Payment card information should preferably be handled by an established payment provider rather than stored on the platform's own servers.

Ownership

The course owner must retain ownership/control of:

- Domain
- Website
- Source code, where applicable
- Database
- Course content
- Video library
- Hosting
- Cloudflare account
- Payment gateway account
- Analytics
- Business email
- Design assets
- Documentation

Developers should receive access to these accounts rather than owning the accounts themselves.

The system should be transferable to another developer if necessary.

Deliverables

The quotation should include, where applicable:

- UI/UX design
- Platform development
- Database setup
- Authentication
- Payment integration
- Video integration
- Course/LMS functionality
- Quiz system
- Progress tracking
- Admin interface
- Responsive design
- Security configuration
- Analytics
- Deployment
- Testing
- Documentation
- Administrator training
- Post-launch support

Clearly identify what is **included, excluded, optional, and recurring**.

Development Plan

The proposed development plan should include:

1) Discovery/requirements
2) Technical architecture
3) UI/UX design
4) Development
5) Payment integration
6) Video integration
7) Course/content integration
8) Testing
9) Security testing
10) Client acceptance testing
11) Deployment
12) Launch
13) Post-launch support

Testing

Before final payment, the platform must be tested for:

- Registration
- Login
- Password recovery
- Successful payments
- Failed payments
- Automatic enrollment
- Course access restrictions
- Video playback
- Unauthorized video access
- Lesson Completion
- Progress tracking
- Quiz functionality
- Mobile responsiveness
- Admin functionality
- Security
- Backups
- Performance
- Browser compatibility

Proposal Requirements

Please provide:

- Technical approach and technology stack
- Development cost and estimated timeline
- Hosting and third-party service requirements
- Recurring costs, including payment and video hosting
- Maintenance, support and warranty terms
- Ownership and source-code handover terms
- Documentation and training included
- Assumptions, exclusions and optional features/costs

Please also flag anything in this brief that you consider:

- Unnecessary or overly complex
- Risky or unnecessarily expensive
- Better handled by an existing service or platform

Important

The goal is not to build the largest possible platform.

The goal is a lean, reliable, secure and professional Version 1 that provides an excellent learning experience and can grow over time.